P and S Waves: Evidence for Earth's Layers

Javalab Seismic Wave · Javalab

Year 8	**~30 minutes**	**Science · Earth and Space Science**

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Simulation: https://javalab.org/en/seismic_wave_en/
Curriculum links: AC9S8U03 · AC9S8U04

Learning intentions

- I can describe how P waves and S waves travel differently through Earth's layers.
- I can explain how seismic wave behaviour provided evidence that Earth's outer core is liquid.
- I can relate wave evidence to the rock cycle and Earth's internal structure.

Getting started

Open the Javalab Seismic Wave simulation and trigger an earthquake. Watch how the P wave and S wave travel outward from the earthquake's origin through Earth's different layers, and pay attention to what happens when each wave reaches the core.

Tasks

1. Trigger an earthquake. Which wave (P or S) reaches a given distance first? What does this tell you about their relative speeds?

2. Watch what happens to the S wave when it reaches the outer core. Describe what you observe.

3. Watch what happens to the P wave when it reaches the outer core. Does it behave the same way as the S wave?

4. Record whether each wave type is able to pass through solid layers and whether it is able to pass through liquid layers.

Wave type	Passes through liquid? (Y/N)	Passes through solid? (Y/N)

5. S waves cannot travel through liquids. Using this fact and what you observed in the simulation, explain how scientists used seismic waves as evidence that Earth's outer core is liquid.

6. Explain why there is a 'shadow zone' on the opposite side of Earth from an earthquake where certain waves cannot be detected.

7. Explain how seismic evidence connects to the rock cycle — for example, why molten material behaves differently to solid rock when a wave passes through it.

8. If scientists only had P wave data (no S wave data), would they still be able to tell that the outer core is liquid? Explain your reasoning.

Extension (optional)

Real seismologists use a network of seismic stations around the world, not just one. Explain why detecting the same earthquake from multiple locations helps them build a more accurate picture of Earth's internal structure.